FreshLien

Same-day foreclosure intelligence — before the crowd

Master SaaS Product & Technical Document
v1.0 — June 2026

$800B+	$99/mo	30–60 Days	Same-Day
Distressed Asset Market	Target Entry Price	Competitor Data Delay	FreshLien Freshness

Prepared for:	Waqas Dost — Founder, freshlien.com
Domain:	freshlien.com
Status:	Go-to-Build MVP Document
Stack:	Supabase (PostgreSQL) · FastAPI · React · Scrapy · Mapbox
Focus:	Pre-Foreclosure · Foreclosure · Probate · Mortgage Default

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1 Product Vision & Market Opportunity

Why FreshLien wins, who it serves, and the size of the prize

What FreshLien Is

FreshLien is a B2B SaaS platform that aggregates, normalises, and delivers same-day distressed real estate intelligence — pre-foreclosure (NOD / Lis Pendens), active foreclosure auction data, probate filings, and mortgage default signals — directly scraped from US county recorders, clerk-of-court portals, and public court databases. The product is sold as a subscription dashboard, a REST API, and bulk data exports to real estate investors, wholesale operators, mortgage servicers, attorneys, and hedge funds.

Core Insight — Speed Is the Product

THE ENTIRE PITCH

PropStream (1M+ users, $99/mo) pulls data through third-party aggregators → 30–60 day lag.
FreshLien scrapes county sources DIRECTLY → same-day or next-day freshness.
Investors who see a pre-foreclosure lead 30 days before competitors can contact the homeowner before anyone else — that first-mover advantage is worth $10,000s per deal.

Data Categories FreshLien Covers

Category	Filing Type	Stage	Lead Value
Pre-Foreclosure	NOD / Lis Pendens	Before auction	HIGHEST — homeowner still in home
Active Foreclosure	NTS / Sheriff Sale / Auction	Scheduled sale	HIGH — time-sensitive
REO / Bank-Owned	Post-auction deed	Lender owns it	MEDIUM — negotiable
Probate	Probate court filing	Estate selling	HIGH — motivated seller
Mortgage Default	Tax lien / HELOC default	Payment missed	HIGH — early warning signal
Tax Delinquency	County tax records	Pre-lien	MEDIUM — slow burn lead

Target Customer Segments

Segment	Pain Point	FreshLien Value	Willingness to Pay
Fix & Flip Investor	Stale data, miss best deals	Same-day NOD alerts	$79–199/mo
Wholesale Investor	Manual research 4–6 hrs/week	165+ filters, bulk export	$199/mo
Mortgage Servicer	No portfolio distress monitor	API flag on any parcel	$999+/mo
Real Estate Attorney	Hours of courthouse search	Full property timeline	$199/mo
Hard Money Lender	Hidden collateral distress	Collateral risk score	$999/mo
Credit Repair Co.	Finding pre-foreclosure leads	Targeted mail lists	$199/mo
Hedge Fund / PE	Nationwide deal sourcing	Full API + custom feed	$2k–10k/mo

Segment	Pain Point	FreshLien Value	Willingness to Pay
Probate Attorney	Finding estate properties	Probate filing alerts	$199/mo

2 Full Database Schema

Supabase (PostgreSQL) — every table, column, index, and relationship

DATABASE CHOICE

MVP: Use Supabase free tier (500 MB, 2 CPU). Production: Supabase Pro ($25/mo) or self-hosted PostgreSQL on AWS RDS (db.t3.micro = ~$15/mo). PostGIS extension required.

Core Tables Overview

Table	Purpose	Row Estimate (Year 1)
properties	Master parcel/property record — one row per address	~500,000
foreclosure_filings	Every NOD, Lis Pendens, NTS, auction filing	~200,000/yr
probate_filings	Probate court filings linked to property	~50,000/yr
tax_liens	Delinquent tax records and municipal liens	~100,000/yr
mortgage_records	Deed of trust, mortgage origination, satisfaction	~300,000/yr
auction_results	Courthouse sale outcomes — sold price, buyer	~80,000/yr
owners	Deduplicated owner/person entity table	~400,000
owner_property_link	Many-to-many owner↔property	~600,000
users	SaaS user accounts	<10,000
subscriptions	Stripe subscription state per user	<10,000
saved_searches	User-defined filter presets	<50,000
alerts	Watch-area alert configurations	<30,000
alert_deliveries	Log of every alert email sent	<500,000/yr
api_keys	API key credentials for B2B tier	<1,000
skip_trace_log	Pay-per-use skip trace requests + responses	<200,000/yr
scrape_jobs	Scraper run history, status, error log	<500,000/yr
counties	Reference table — all 3,143 US counties	3,143

SQL: Core Table Definitions

SQL — Supabase / PostgreSQL

```
-- Enable PostGIS & UUID
CREATE EXTENSION IF NOT EXISTS postgis;
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

-- COUNTIES reference table
CREATE TABLE counties (
id SERIAL PRIMARY KEY,
```

```
fips CHAR(5) UNIQUE NOT NULL, -- e.g. '12086' = Miami-Dade
name TEXT NOT NULL,
state CHAR(2) NOT NULL, -- e.g. 'FL'
state_full TEXT,
judicial BOOLEAN DEFAULT FALSE, -- TRUE = Lis Pendens state
scraper_url TEXT, -- county portal base URL
active BOOLEAN DEFAULT TRUE
);

-- PROPERTIES master parcel table
CREATE TABLE properties (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
parcel_id TEXT, -- county APN/parcel number
county_fips CHAR(5) REFERENCES counties(fips),
address_full TEXT NOT NULL,
address_street TEXT,
address_city TEXT,
address_state CHAR(2),
address_zip TEXT,
property_type TEXT, -- SFR, MFR, CONDO, LAND, COMM
bedrooms INT,
bathrooms NUMERIC(3,1),
sqft INT,
lot_sqft INT,
year_built INT,
assessed_value NUMERIC(12,2),
last_sale_price NUMERIC(12,2),
last_sale_date DATE,
estimated_value NUMERIC(12,2), -- AVM estimate
estimated_equity NUMERIC(12,2), -- est_value - est_balance
tax_status TEXT, -- CURRENT, DELINQUENT, EXEMPT
tax_annual NUMERIC(10,2),
flood_zone TEXT, -- FEMA zone code e.g. AE, X
zoning TEXT,
geom GEOMETRY(Point, 4326), -- PostGIS lat/lng
latitude NUMERIC(10,7),
longitude NUMERIC(10,7),
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_properties_zip ON properties(address_zip);
CREATE INDEX idx_properties_county ON properties(county_fips);
CREATE INDEX idx_properties_geom ON properties USING GIST(geom);

-- FORECLOSURE FILINGS (core product data)
CREATE TABLE foreclosure_filings (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
property_id UUID REFERENCES properties(id),
county_fips CHAR(5) REFERENCES counties(fips),
filing_type TEXT NOT NULL, -- NOD, LIS_PENDENS, NTS, AUCTION, REO
filing_stage TEXT, -- PRE_FORECLOSURE, ACTIVE, POST
case_number TEXT,
filing_date DATE NOT NULL,
recording_date DATE,
auction_date DATE,
judgment_amount NUMERIC(12,2), -- from sample data: Approx Judgment
```

```
default_amount NUMERIC(12,2),
lender_name TEXT,
lender_type TEXT, -- BANK, CREDIT_UNION, PRIVATE, GOV
trustee_name TEXT,
attorney_name TEXT,
-- Defendants (from sample CSV)
defendants TEXT[], -- array of defendant names
defendant_primary TEXT, -- first named defendant
-- Additional lien holders
junior_liens TEXT[], -- HOA, IRS, STATE, etc.
federal_lien BOOLEAN DEFAULT FALSE,-- USA as defendant = federal lien
state_lien BOOLEAN DEFAULT FALSE,
hoa_lien BOOLEAN DEFAULT FALSE,
-- Computed fields
days_to_auction INT, -- computed from auction_date
days_since_filing INT, -- computed from filing_date
-- Source tracking
source_url TEXT,
source_portal TEXT,
property_id_raw TEXT, -- from sample: Property ID col
scrape_job_id UUID,
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_ff_property ON foreclosure_filings(property_id);
CREATE INDEX idx_ff_county ON foreclosure_filings(county_fips);
CREATE INDEX idx_ff_filing_date ON foreclosure_filings(filing_date DESC);
CREATE INDEX idx_ff_auction_date ON foreclosure_filings(auction_date);
CREATE INDEX idx_ff_type ON foreclosure_filings(filing_type);
```

```
-- PROBATE FILINGS
CREATE TABLE probate_filings (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
property_id UUID REFERENCES properties(id),
county_fips CHAR(5) REFERENCES counties(fips),
case_number TEXT,
decedent_name TEXT,
estate_value NUMERIC(12,2),
filing_date DATE,
hearing_date DATE,
attorney TEXT,
executor TEXT,
status TEXT, -- OPEN, CLOSED, CONTESTED
source_url TEXT,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

```
-- TAX LIENS
CREATE TABLE tax_liens (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
property_id UUID REFERENCES properties(id),
county_fips CHAR(5),
lien_type TEXT, -- PROPERTY_TAX, MUNICIPAL, IRS
lien_amount NUMERIC(12,2),
years_delinquent INT,
filing_date DATE,
```

```
created_at TIMESTAMPTZ DEFAULT NOW()
);

-- OWNERS (deduplicated entity)
CREATE TABLE owners (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
full_name TEXT,
name_variants TEXT[], -- AKAs from court docs
mailing_address TEXT,
-- skip trace appended fields
phone_numbers TEXT[],
emails TEXT[],
skip_traced_at TIMESTAMPTZ,
created_at TIMESTAMPTZ DEFAULT NOW()
);

-- USERS & SUBSCRIPTIONS
CREATE TABLE users (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
email TEXT UNIQUE NOT NULL,
name TEXT,
company TEXT,
role TEXT DEFAULT 'investor',-- investor, servicer, attorney, lender
created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE subscriptions (
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
user_id UUID REFERENCES users(id),
stripe_customer_id TEXT,
stripe_sub_id TEXT,
plan TEXT NOT NULL, -- STARTER, PRO, ENTERPRISE
status TEXT, -- active, trialing, canceled
states_allowed TEXT[], -- e.g. ['FL','TX','CA']
api_calls_limit INT,
views_limit INT,
current_period_start DATE,
current_period_end DATE,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

3 Data Sources

Every source, URL pattern, filing type, cost, and priority

Source Priority Matrix

Source	Data Type	Cost	Freshness	Priority
County Recorder Portals	NOD, NTS, Deed of Trust	Free	Same-day	P0 — Build first
Clerk of Court (Judicial)	Lis Pendens, foreclosure case	Free	Same-day	P0 — FL/NJ/NY
County Assessor / Tax	APN, value, owner, tax status	Free	Weekly	P0 — enrich all
PACER (Federal Courts)	Bankruptcy Ch.7 / Ch.13	$0.10/pg	Same-day	P1 — cross-ref
HUD.gov REO	Bank-owned listings	Free	Weekly	P1 — REO stage
FHFA (Fannie/Freddie)	GSE foreclosure data	Free	Monthly	P1 — portfolio
Probate Court Portals	Estate filings	Free	Weekly	P1 — probate tier
ATTOM API (gap fill)	Any county, all data types	$200–500/mo	Daily	P2 — rural gaps
FEMA Flood Maps API	Flood zone per parcel	Free	Annual	P1 — risk layer
Census TIGER / Zoning	Parcel geometry, zoning	Free	Annual	P1 — geo layer
BatchSkipTracing API	Phone, email per owner	$0.05–0.15/rec	On-demand	P2 — add-on

Sample Data Structure — Burlington County NJ Format

The uploaded sample data shows the exact format scraped from Burlington County, NJ. Every filing row maps to these fields in the foreclosure_filings and properties tables:

CSV Column	DB Table	DB Column	Notes
Property ID	foreclosure_filings	property_id_raw	Court docket / internal ID
Address	properties	address_full	Parse into street/city/state/zip
Defendant	owners + owner_property_link	full_name / name_variants	Split on semicolons; flag AKAs
Sales Date	foreclosure_filings	auction_date	The scheduled sheriff/court sale
Approx Judgment	foreclosure_filings	judgment_amount	Parse currency → NUMERIC
(derived) Snapshot Date	scrape_jobs	run_date	Header row date from source
(derived) State	properties	address_state	Parse from address string
(derived) County	properties / ff	county_fips	Lookup from county name
(derived) Federal Lien	foreclosure_filings	federal_lien	TRUE if 'UNITED STATES OF AMERICA' in d
(derived) State Lien	foreclosure_filings	state_lien	TRUE if 'STATE OF NEW JERSEY' etc.
(derived) HOA Lien	foreclosure_filings	hoa_lien	TRUE if 'ASSOCIATION' in defendants

SAMPLE DATA NOTES

The NJ Burlington County sample has 19 filings across two auction dates (11/20/2025 & 12/4/2025).

Judgment amounts range from $35K to $494K. Multiple defendants per filing (semicolon-delimited).

Federal liens (USA), state liens (NJ), and HOA liens appear in the defendants column — always parse these.

4 Scraper Architecture & ETL Pipeline

Scrapy + Selenium → S3 → normalise → Supabase — full spec

Architecture Overview

Each county gets its own Scrapy spider. Spiders are scheduled by Apache Airflow (or a lightweight cron on EC2) to run at 6:00 AM EST daily. Raw HTML/JSON is archived to S3. The ETL normaliser reads from S3, parses fields, geocodes addresses, detects lien types, and upserts into Supabase via the PostgREST API.

File Structure

PROJECT STRUCTURE

```
freshlien-scraper/
■■■ spiders/
■ ■■■ base_spider.py # shared auth, retry, CAPTCHA logic
■ ■■■ fl_miami_dade.py # Florida — Miami-Dade Clerk of Court
■ ■■■ fl_broward.py
■ ■■■ fl_palm_beach.py
■ ■■■ tx_harris.py # Texas — Harris County (Houston)
■ ■■■ tx_dallas.py
■ ■■■ ca_la.py # California — LA County Recorder
■ ■■■ nj_burlington.py # NJ Burlington (matches sample data)
■ ■■■ ... (30 counties MVP)
■■■ etl/
■ ■■■ normaliser.py # field parsing, address splitting
■ ■■■ geocoder.py # Google Maps or Nominatim
■ ■■■ lien_detector.py # parse defendants → lien flags
■ ■■■ upsert.py # Supabase PostgREST upsert
■ ■■■ assessor_enricher.py # merge county assessor data
■■■ dags/
■ ■■■ daily_scrape.py # Airflow DAG
■■■ settings.py # Scrapy settings
■■■ requirements.txt
```

NJ Burlington Scraper — Concrete Example

spiders/nj_burlington.py

```
import scrapy, re
from .base_spider import BaseForeclosureSpider
from etl.normaliser import normalise_filing
from etl.lien_detector import detect_liens

class NJBurlingtonSpider(BaseForeclosureSpider):
name = 'nj_burlington'
county_fips = '34005'
base_url = 'https://www.burlingtoncountynj.gov/sheriff/foreclosure'

def parse(self, response):
for row in response.css('table.foreclosure-list tr:not(:first-child)'):
cols = row.css('td::text').getall()
if len(cols) < 5: continue
raw = {
'property_id_raw': cols[0].strip(),
'address_full': cols[1].strip(),
```

```
'defendants_raw': cols[2].strip(),
'auction_date': cols[3].strip(),
'judgment_amount': cols[4].strip(),
'county_fips': self.county_fips,
'filing_type': 'LIS_PENDENS',
'source_url': response.url,
}
filing = normalise_filing(raw)
filing.update(detect_liens(raw['defendants_raw']))
yield filing
```

Lien Detector Logic

etl/lien_detector.py

```
import re

FEDERAL_KEYWORDS = ['UNITED STATES OF AMERICA', 'U.S.A.', 'IRS',
'SMALL BUSINESS ADMINISTRATION']
STATE_KEYWORDS = ['STATE OF NEW JERSEY', 'STATE OF FLORIDA',
'NJHMFA', 'NEW JERSEY HOUSING AND MORTGAGE']
HOA_KEYWORDS = ['ASSOCIATION', 'HOA', 'CONDOMINIUM', 'COMMUNITY']
PROBATE_KEYWORDS = ['ESTATE OF', 'EXECUTRIX', 'ADMINISTRATOR OF']

def detect_liens(defendants_raw: str) -> dict:
d = defendants_raw.upper()
names = [n.strip() for n in defendants_raw.split(';')]
return {
'defendants': names,
'defendant_primary': names[0] if names else None,
'junior_liens': [n for n in names if any(k in n.upper()
for k in HOA_KEYWORDS+STATE_KEYWORDS+FEDERAL_KEYWORDS)],
'federal_lien': any(k in d for k in FEDERAL_KEYWORDS),
'state_lien': any(k in d for k in STATE_KEYWORDS),
'hoa_lien': any(k in d for k in HOA_KEYWORDS),
'probate_flag': any(k in d for k in PROBATE_KEYWORDS),
}
```

5 Backend API — FastAPI Specification

Every endpoint, request/response schema, and auth rule

Technology Stack

FastAPI (Python 3.11) · Supabase (PostgreSQL + PostGIS) · Redis (hot cache) · Stripe (billing) · SendGrid (email alerts) · Resend (transactional) · JWT (supabase-py auth) · Railway or Render for free-tier hosting initially.

API Endpoints

Method	Endpoint	Auth	Description
GET	/api/v1/foreclosures	JWT	List/search filings with 50+ filters
GET	/api/v1/foreclosures/{id}	JWT	Full property detail with timeline
GET	/api/v1/properties/{id}	JWT	Property record + all linked filings
GET	/api/v1/search	JWT	Full-text + geo search
GET	/api/v1/map/clusters	JWT	GeoJSON clusters for map view
GET	/api/v1/stats/county/{fips}	JWT	County-level filing stats
POST	/api/v1/alerts	JWT	Create watch-area alert
GET	/api/v1/alerts	JWT	List user's alerts
DELETE	/api/v1/alerts/{id}	JWT	Delete alert
POST	/api/v1/export/csv	JWT	Export filtered results as CSV
POST	/api/v1/skip-trace	JWT+PRO	Append phone/email to owner
GET	/api/v1/risk-score/{prop_id}	JWT	Collateral risk score (0–100)
GET	/api/v1/portfolio/monitor	JWT+ENT	Portfolio distress check
POST	/api/v1/api-keys	JWT+ENT	Generate B2B API key
GET	/api/v1/usage	JWT	API call usage this billing period
POST	/api/v1/webhooks/stripe	Stripe-sig	Stripe webhook handler

Key Query Parameters — /api/v1/foreclosures

Parameter	Type	Example	Description
state	str[]	FL,TX,CA	Filter by state(s)
county_fips	str[]	12086,48201	Filter by FIPS code
zip	str[]	33101,77002	Filter by ZIP code
filing_type	enum	NOD,LIS_PENDENS	Filing category
filing_stage	enum	PRE_FORECLOSURE	Stage of process

Parameter	Type	Example	Description
days_since_filing_max	int	30	Filed within N days
auction_date_from	date	2025-11-01	Auction date range start
auction_date_to	date	2025-12-31	Auction date range end
judgment_min	float	50000	Min judgment amount
judgment_max	float	500000	Max judgment amount
equity_min_pct	float	20	Min estimated equity %
federal_lien	bool	false	Exclude/include federal liens
property_type	enum	SFR,CONDO	Property type filter
has_auction_date	bool	true	Only filings with scheduled sale
probate	bool	true	Include probate-flagged filings
sort	str	filing_date:desc	Sort field and direction
limit	int	50	Page size (max 200 Starter, unlimited Pro)
offset	int	0	Pagination offset

FastAPI Code — Main Application

main.py — FastAPI

```
# main.py
from fastapi import FastAPI, Depends, HTTPException
from fastapi.middleware.cors import CORSMiddleware
from supabase import create_client
import redis, os

app = FastAPI(title='FreshLien API', version='1.0')
app.add_middleware(CORSMiddleware, allow_origins=['*'],
allow_methods=['*'], allow_headers=['*'])

supabase = create_client(os.environ['SUPABASE_URL'],
os.environ['SUPABASE_SERVICE_KEY'])
cache = redis.Redis.from_url(os.environ['REDIS_URL'])

@app.get('/api/v1/foreclosures')
async def list_foreclosures(
state: str = None, county_fips: str = None,
zip: str = None, filing_type: str = None,
judgment_min: float = None, judgment_max: float = None,
days_since_filing_max: int = None,
federal_lien: bool = None,
sort: str = 'filing_date:desc',
limit: int = 50, offset: int = 0,
user = Depends(get_current_user)
):
cache_key = f'ff:{state}:{zip}:{filing_type}:{limit}:{offset}'
cached = cache.get(cache_key)
if cached: return json.loads(cached)

q = supabase.table('foreclosure_filings_view').select('*')
if state: q = q.in_('address_state', state.split(','))
if county_fips: q = q.in_('county_fips', county_fips.split(','))
if zip: q = q.in_('address_zip', zip.split(','))
if filing_type: q = q.in_('filing_type', filing_type.split(','))
if judgment_min: q = q.gte('judgment_amount', judgment_min)
if judgment_max: q = q.lte('judgment_amount', judgment_max)
if federal_lien is not None: q = q.eq('federal_lien', federal_lien)
if days_since_filing_max:
from datetime import date, timedelta
cutoff = date.today() - timedelta(days=days_since_filing_max)
q = q.gte('filing_date', str(cutoff))
sort_col, sort_dir = (sort+':desc').split(':')[:2]
q = q.order(sort_col, desc=(sort_dir=='desc'))
q = q.range(offset, offset + limit - 1)
result = q.execute()
cache.setex(cache_key, 300, json.dumps(result.data))
return result.data
```

6 Frontend Dashboard

React + Mapbox + Tailwind — every screen, component, and interaction

Tech Stack

React 18 · Vite · Tailwind CSS · shadcn/ui components · Mapbox GL JS · React Query (TanStack) · Zustand (state) · Recharts (analytics) · Supabase Auth UI. Deploy: Vercel (free tier for MVP).

Application Routes

Route	Page / Component	Access Level
/	Landing page (marketing)	Public
/login	Auth — email/password + Google OAuth	Public
/dashboard	Map view + filter sidebar (main product)	Starter+
/dashboard/list	Table/list view of filings	Starter+
/dashboard/property/:id	Property detail drawer/page	Starter+
/dashboard/alerts	Manage watch-area alerts	Starter+
/dashboard/saved	Saved searches	Starter+
/dashboard/exports	Export history + download	Starter+
/dashboard/skip-trace	Skip trace credits & history	Pro+
/dashboard/api	API key management	Enterprise
/dashboard/portfolio	Portfolio monitor	Enterprise
/analytics	Market analytics & charts	Pro+
/settings	Account + billing + subscription	All
/admin	Internal data ops dashboard	Admin only

Dashboard Map View — Component Spec

The map view is the core product screen. It has three zones:

Zone	Component	Description
Left sidebar (320px)	FilterPanel	All search filters: state, county, ZIP, type, date, judgment range, equity, lien f
Center (flex)	MapView (Mapbox GL)	Clustered pins for all matching filings. Click pin → PropertyDrawer slides in f
Right drawer (420px)	PropertyDrawer	Full detail: address, owner, judgment, auction date, lien holders, equity estim
Top bar	SearchBar + ViewToggle	Address search, map/list toggle, active filter tags, result count, export CSV b
Bottom bar	ResultStats	Total results, avg judgment, auction calendar strip

Map Pin Color Coding

Color	Hex	Meaning
Red	#E63946	Auction in < 7 days — URGENT
Orange	#F4A261	Auction in 7–30 days
Yellow	#FFD166	Auction in 30–90 days
Blue	#00B4D8	Pre-foreclosure — no auction date yet
Purple	#7B2D8B	Probate filing
Gray	#94A3B8	REO / post-auction

7 AI / RAG Layer

Natural language search + property risk reports using Claude / GPT-4o

Features

Feature	Description	Model	Tier
Natural Language Search	"Show 3-bed pre-foreclosures in Miami with 50%+ equity, no flood risk"	Claude Sonnet / GPT-4o	Pro+
Property Risk Report	Plain-English explanation of all liens, risks, auction timeline	GPT-4o	$4.99/report
Market Summary	"What's happening in ZIP 33101 this week?" — auto-generated	Claude Haiku	Pro+
Document Parser	Upload a lien document PDF → extract fields automatically	Claude Sonnet	Enterprise
Distress Score Explanation	Explain why a property scored 87/100 in plain English	GPT-4o	Pro+

NL Search Prompt Template

AI PROMPT — NL SEARCH PARSER

```
SYSTEM: You are FreshLien's query parser. Convert natural language queries
into a structured JSON filter object. Valid fields: state (2-char array),
county_fips (array), zip (array), filing_type (NOD|LIS_PENDENS|NTS|AUCTION|REO),
property_type (SFR|CONDO|MFR|LAND), bedrooms_min, equity_min_pct,
judgment_min, judgment_max, federal_lien (bool), flood_zone_exclude (bool),
days_since_filing_max, has_auction_date (bool).
Respond ONLY with valid JSON, no explanation.

USER: Show me 3-bed pre-foreclosures in Miami with 50%+ equity and no flood risk

RESPONSE:
{
"state": ["FL"],
"county_fips": ["12086"],
"filing_type": "LIS_PENDENS",
"bedrooms_min": 3,
"equity_min_pct": 50,
"flood_zone_exclude": true
}
```

8 Authentication, Subscriptions & Billing

Supabase Auth + Stripe — complete implementation notes

Auth Flow

Use Supabase Auth (built-in). Email/password + Google OAuth. On sign-up, create a users row and a subscriptions row with plan=TRIAL (7-day free). JWT is issued by Supabase and validated on every FastAPI request via supabase-py.

Stripe Integration

- Create 3 Stripe Products: Starter ($79/mo), Pro ($199/mo), Enterprise ($999/mo)
- Create Stripe Checkout Session on plan selection → redirect to Stripe hosted page
- Webhook: stripe.checkout.session.completed → set subscriptions.status='active', plan=X
- Webhook: customer.subscription.deleted → set status='canceled', block API access
- Add-on: skip_trace charges → Stripe metered billing or one-time PaymentIntent ($0.25/rec)
- Show current usage vs limits in /settings page (views used, API calls, skip traces)

Rate Limiting by Plan

Feature	Starter ($79)	Pro ($199)	Enterprise ($999)
States	1	5	All 50
Property views/mo	500	Unlimited	Unlimited
CSV export rows	200/export	5,000/export	Unlimited
Alert watch areas	1 ZIP	10 ZIPs	Unlimited polygons
API calls/mo	None	1,000	Unlimited
Skip trace	Not available	$0.25/rec	$0.20/rec (bulk)
AI property reports	Not available	$4.99/report	$2.99/report
Portfolio monitor	Not available	Not available	Included
SLA / support	Email	Priority email	Dedicated Slack

9 Pricing Model & Revenue Projections

3-tier SaaS + high-margin add-ons — 24-month model

Revenue Projections (Conservative)

Revenue Line	Month 3	Month 6	Month 12	Month 24
Starter @$79 (users)	$790 (10)	$3,950 (50)	$11,850 (150)	$31,600 (400)
Pro @$199 (users)	$995 (5)	$5,970 (30)	$19,900 (100)	$59,700 (300)
Enterprise @$999 (clients)	$0	$1,998 (2)	$9,990 (10)	$29,970 (30)
Skip Trace add-on	$200	$800	$3,000	$9,000
AI Reports add-on	$0	$500	$2,000	$6,000
Direct mail lists	$0	$300	$1,000	$3,000
TOTAL MRR	~$2k	~$14k	~$48k	~$139k
TOTAL ARR	~$24k	~$168k	~$576k	~$1.67M

Monthly Operating Costs

Expense	Month 1–3	Month 6+	Notes
Supabase	$0	$25	Free tier MVP → Pro at scale
AWS EC2 (scrapers)	$30	$120	t3.medium for 30 spiders
AWS S3 (raw archive)	$5	$20	Cheap storage
Redis (Upstash free)	$0	$20	Free tier → paid at scale
ATTOM API gap fill	$200	$200	Reduce as coverage grows
Google Maps Geocoding	$0	$50	Use Nominatim free first
LLM API (Claude/GPT)	$20	$100	AI features only
Stripe fees	$60	$420	2.9% + $0.30 per transaction
SendGrid / Resend	$0	$20	Free tier covers launch
Domain + tools	$15	$15	freshlien.com + Cloudflare
TOTAL COSTS	~$330	~$970	~92% net margin at Month 6

10 Competitive Analysis

How FreshLien beats every competitor on the only metric that matters: freshness

Competitor	Price	Data Lag	Probate	API	FreshLien Edge
PropStream	$99/mo	30–60 days	No	Limited	Same-day + probate + API
BatchData	$500/mo	7–14 days	No	Yes	10x cheaper, has UI
ATTOM Data	$5k–50k/yr	3–7 days	Partial	Yes	20–100x cheaper
PropertyShark	$150–275/mo	3–7 days	No	No	API + probate + cheaper
CoreLogic	$10k+/yr	1–3 days	No	Yes	Affordable for SMB
REDX	$70–150/mo	7–30 days	No	No	Fresher + richer data
FreshLien	$79–999/mo	SAME-DAY	YES	YES	— THE BENCHMARK —

CORE MOAT

The entire competitive moat = county-direct scraping → same-day data → investor contacts homeowner BEFORE PropStream users even see the filing. That's the pitch. Never dilute it.

11

60-Day MVP Build Scope

Smallest shippable product that investors will pay for

Feature	In MVP?	Notes
3 states (FL, TX, CA) + NJ Burlington (sample)	YES	30 county scrapers
Supabase DB (free tier)	YES	No PostGIS needed for MVP
FastAPI backend on Render.com (free)	YES	Deploy in 30 min
React dashboard with Mapbox map	YES	Vercel free deploy
Filter panel (state, ZIP, type, date, judgment)	YES	Core product
Property detail drawer	YES	All scraped fields
CSV export (200 rows — Starter limit)	YES	Papa Parse
Email alerts (1 ZIP — Starter)	YES	SendGrid cron
Stripe subscriptions (Starter + Pro)	YES	Launch day billing
Skip trace add-on	YES	BatchSkipTracing API
AI natural language search	NO	v2 feature
Portfolio monitoring	NO	Enterprise v2
All 50 states	NO	Add 5 states/sprint
Probate court scraping	NO	Planned v1.5
Mobile app	NO	Not needed — responsive web

12

90-Day Launch Roadmap

Week-by-week build, validate, and launch plan

Week	Goal	Tasks
1–2	Validate demand	Post in 3 RE investor FB groups. Get 20 waitlist signups. Carrd landing page.
3–5	Data pipeline — FL	Scrapers for Miami-Dade, Broward, Palm Beach. ETL → Supabase. 500 rows in DB.
6–7	MVP dashboard	React + Mapbox. Filter sidebar. Property drawer. Vercel deploy.
8	TX + CA scrapers	Harris (Houston), Dallas. LA, Riverside. Now 3 states live.
9	NJ scraper (sample)	Burlington County spider. Matches your sample data format exactly.
10	Auth + Billing	Supabase Auth. Stripe Starter + Pro plans. Email alerts (SendGrid).
11	Beta users	10 free beta users from waitlist. 30-min Zoom each. Collect feedback.
12	Fix + Launch	Fix top 3 complaints. Get 3 testimonials. ProductHunt launch. BiggerPockets post.
13+	Growth sprint	Add 3 states/mo. Add probate tier. Launch skip trace add-on. API tier.

13 Infrastructure & DevOps

Free-tier MVP → production scale path

MVP Stack (Free / Near-Free)

Service	Used For	Free Tier	Paid Trigger
Supabase	PostgreSQL + Auth + Storage	500 MB, 2 CPU	>500 MB or >50k rows/hr
Render.com	FastAPI backend	750 hrs/mo free	>750 hrs or need >256 MB RAM
Vercel	React frontend	Unlimited deploys	Team features
Upstash Redis	API response cache	10k req/day	>10k req/day
SendGrid	Alert emails	100 emails/day	>100/day or custom domain
GitHub Actions	CI/CD	2000 min/mo	Rarely exceeded
Cloudflare	CDN + SSL + DNS	Free forever	Only for advanced features

Scraper Hosting

Scrapers run on AWS EC2 t3.micro ($8/mo) or a cheap VPS (Hetzner €4/mo). Airflow runs locally or on the same instance via systemd cron. Raw scraped data is archived to S3 ($1–2/mo). Use GitHub Actions to auto-deploy new spider code on push to main.

Environment Variables

.env — Environment Variables

```
# .env — keep this file OUT of git (add to .gitignore)
SUPABASE_URL=https://xxxx.supabase.co
SUPABASE_SERVICE_KEY=eyJhbGci... # service role key (server only)
SUPABASE_ANON_KEY=eyJhbGci... # public anon key (frontend)
REDIS_URL=redis://default:xxx@upstash.io:6379
STRIPE_SECRET_KEY=sk_live_xxx
STRIPE_WEBHOOK_SECRET=whsec_xxx
STRIPE_STARTER_PRICE_ID=price_xxx
STRIPE_PRO_PRICE_ID=price_xxx
STRIPE_ENT_PRICE_ID=price_xxx
SENDGRID_API_KEY=SG.xxx
MAPBOX_TOKEN=pk.eyJ1... # public — safe in frontend
OPENAI_API_KEY=sk-xxx # or ANTHROPIC_API_KEY
ANTHROPIC_API_KEY=sk-ant-xxx
ATTOM_API_KEY=xxx # gap-fill data
BATCH_SKIP_TRACE_KEY=xxx
TWOCAPTCHA_KEY=xxx # CAPTCHA solving for scrapers
AWS_ACCESS_KEY_ID=xxx
AWS_SECRET_ACCESS_KEY=xxx
S3_BUCKET=freshlien-raw-filings
GEOCODING_PROVIDER=nominatim # or 'google'
GOOGLE_MAPS_KEY=xxx # only if using Google geocoding
ENVIRONMENT=development # or 'production'
```

14 Legal, Compliance & Data Privacy

Public record = legal to scrape, but know the rules

- All foreclosure, probate, and court filings are PUBLIC RECORD under US law. You have the right to access and republish them.
- The FCRA (Fair Credit Reporting Act) applies if you sell data for credit, insurance, or employment decisions. FreshLien sells for REAL ESTATE investment only — not FCRA-regulated.
- GLBA (Gramm-Leach-Bliley) does NOT apply — you are not a financial institution.
- Skip trace data: obtain ONLY from FCRA-compliant providers (TLO, IDI, BatchSkipTracing all certify this). Your TOS must require users to use skip trace data lawfully.
- GDPR / international: Your customers are US-based. Your servers should be in the US. Minimal GDPR exposure for MVP.
- CAN-SPAM: All alert and marketing emails must include an unsubscribe link. Use SendGrid's built-in compliance tools.
- Terms of Service for each county portal: Most prohibit commercial resale. Mitigate by: (a) only displaying derived data, not raw portal HTML; (b) staying below aggressive request rates (1 req/sec max); (c) respecting robots.txt.
- Incorporate as LLC in Wyoming or Delaware (~$100/yr). Opens US bank account. Use Mercury Bank (free, online, Pakistan-friendly).
- Legal pages required at launch: Privacy Policy, Terms of Service, Data Use Policy. Use Termly.io ($10/mo) to auto-generate.

15 Go-to-Market Strategy

How to get your first 100 paying customers from Pakistan

Channel	Action	Expected Result	Cost
BiggerPockets Forum	Weekly posts sharing county data insights	100–500 visitors/post	Free
Facebook Groups	"Real Estate Investing" groups (1M+ members) — share free data snapshots	50–200 signups/post	Free
Reddit r/realestateinvesting	Share NJ/FL/TX data insights, tools AMA	500–2k views/post	Free
YouTube SEO	"How to find pre-foreclosures in [city] before anyone else"	Long-tail organic traffic	Time only
Direct outreach	Email 100 active real estate wholesalers — 10% response	5–10 trials	Free
ProductHunt	Launch day — prep 2 weeks in advance	500–2k visits on launch day	$0
Paid Meta ads	Target: US, real estate investor interests, lookalike from email list	$500/mo → 20–50 trials	$500/mo
Affiliate program	Pay $20/referral to real estate educators & YouTubers	Leveraged growth	Revenue share

LAUNCH STRATEGY

First 30 days: post ONLY in BiggerPockets and Facebook groups. Share actual data (e.g. '47 new Lis Pendens filed in Miami-Dade this week — here are the top 5 by equity'). This IS the marketing. Build credibility with free data → convert 10% to paid. No ads until Month 3.

16 Financial Projections — 24 Month

Conservative, base, and stretch scenarios

Metric	Month 1	Month 3	Month 6	Month 12	Month 18	Month 24
Total paying users	5	25	100	350	650	1,100
MRR (blended)	$400	$2,000	$13,800	$47,000	$88,000	$138,000
ARR	$4,800	$24k	$166k	$564k	$1.06M	$1.66M
Operating costs	$330	$500	$970	$2,500	$4,000	$6,000
Net profit/mo	$70	$1,500	$12,800	$44,500	$84,000	$132,000
Net margin	17%	75%	93%	95%	95%	96%
States covered	4	4	10	25	40	50
County scrapers	30	30	80	200	400	600

THE MATH

At Month 12 with 350 users and $47k MRR, you are earning ~$44k net/month.

That is $528k/year net profit from a $330 upfront investment. This is achievable.

Even at 50% of projections = $264k/year. Still life-changing from Pakistan.

17 Cursor / Claude Code Prompt Templates

Copy-paste these into Cursor, Claude Code, or any AI IDE to build each module

Prompt: Supabase Schema

CURSOR / CLAUDE CODE — SUPABASE SCHEMA

```
Create a Supabase PostgreSQL schema for a foreclosure data SaaS called FreshLien.
Include tables: properties, foreclosure_filings, probate_filings, tax_liens, owners,
users, subscriptions, alerts, api_keys, scrape_jobs.
Enable PostGIS extension. Add GiST spatial index on properties.geom.
foreclosure_filings must have: filing_type (NOD/LIS_PENDENS/NTS/AUCTION/REO),
judgment_amount, auction_date, defendants TEXT[], federal_lien BOOL, state_lien BOOL,
hoa_lien BOOL, days_to_auction computed column.
Use UUID primary keys everywhere. Add updated_at trigger on all tables.
```

Prompt: FastAPI Backend

CURSOR / CLAUDE CODE — FASTAPI BACKEND

```
Build a FastAPI application for FreshLien foreclosure data SaaS.
POST /auth endpoints using Supabase Auth (supabase-py).
GET /api/v1/foreclosures with filters: state[], county_fips[], zip[], filing_type,
judgment_min/max, days_since_filing_max, federal_lien bool, sort, limit, offset.
GET /api/v1/foreclosures/{id} returns full property detail joined with owners.
POST /api/v1/alerts creates a watch-area alert (polygon or ZIP).
POST /api/v1/export/csv returns streamed CSV of filtered results.
Add Redis caching (5 min TTL) for list queries.
Add Stripe webhook handler for subscription lifecycle.
Rate-limit by subscription plan (Starter 500 views, Pro unlimited).
```

Prompt: React Dashboard

CURSOR / CLAUDE CODE — REACT DASHBOARD

```
Build a React 18 + Tailwind + shadcn/ui foreclosure intelligence dashboard.
Left sidebar: filter panel with state multiselect, ZIP input, filing_type checkboxes,
judgment range slider ($0–$1M), days_since_filing slider, federal_lien toggle.
Center: Mapbox GL JS map, cluster pins by filing count, color-code by urgency:
red=auction<7days, orange=auction 7-30days, blue=pre-foreclosure no auction date.
Right drawer: slides in on pin click. Shows address, defendant names, judgment amount,
auction date, days until auction, lien holders list (HOA/Federal/State badges),
estimated equity, flood zone badge, skip trace button (Pro only), export row button.
Top bar: address search (geocode), map/list toggle, filter tag chips, result count.
Use React Query for data fetching. Zustand for filter state.
```

Prompt: Burlington NJ Scraper

CURSOR / CLAUDE CODE — BURLINGTON NJ SCRAPER

```
Write a Scrapy spider for Burlington County NJ foreclosure listings.
The source page lists properties with columns: Property ID, Address, Defendant, Sales Date, Approx Judgment.
Defendants are semicolon-delimited. Addresses include alternate mailing addresses.
Parse judgment amounts to float. Parse sales date to ISO date.
Split defendants array. Detect: federal_lien (UNITED STATES OF AMERICA),
```

```
state_lien (STATE OF NEW JERSEY), hoa_lien (ASSOCIATION/CONDOMINIUM in name),
probate_flag (ESTATE OF/EXECUTRIX in name).
Upsert to Supabase via PostgREST API. Log run to scrape_jobs table.
Handle pagination if multiple pages. Retry on 429/503. Archive raw HTML to S3.
```

Prompt: Email Alert System

CURSOR / CLAUDE CODE — EMAIL ALERT SYSTEM

```
Build a FastAPI background task that sends email alerts for FreshLien.
Every 6 hours, for each alert in the alerts table:
1. Query foreclosure_filings for new filings within the alert's ZIP/polygon
that were filed since the last alert delivery.
2. If any new filings found, send email via SendGrid with a table showing:
address, filing type, judgment amount, auction date, days until auction.
3. Log delivery to alert_deliveries table.
Use APScheduler for the background job. Template the email in HTML.
Include unsubscribe link (CAN-SPAM compliance).
```

18 Config & Quick Reference

All URLs, keys, commands, and links you need on day one

Key URLs & Services

Service	URL / Command	Purpose
Supabase dashboard	app.supabase.com	DB, auth, storage management
Vercel deploy	vercel deploy --prod	Frontend deploy CLI
Render deploy	git push → auto-deploy	Backend FastAPI hosting
Stripe dashboard	dashboard.stripe.com	Subscriptions, webhooks, payouts
SendGrid	app.sendgrid.com	Email alerts setup
Upstash Redis	console.upstash.com	Redis cache (free tier)
Mapbox Studio	studio.mapbox.com	Map style + access token
Namecheap	namecheap.com	freshlien.com domain management
Cloudflare	cloudflare.com	DNS + free SSL + CDN
BatchSkipTracing	batchskiptracing.com	Skip trace API for owners
ATTOM API	api.attomdata.com	County gap-fill data
2Captcha	2captcha.com	CAPTCHA solving for scrapers
Mercury Bank	mercury.com	US business bank account
Wyoming LLC	wyomingagents.com	Entity formation ~$100/yr
ProductHunt	producthunt.com/posts/new	Launch day listing
BiggerPockets	biggerpockets.com/forums	Primary GTM channel

Quick-Start Commands

QUICK START COMMANDS

```
# 1. Clone & setup
git clone https://github.com/your-org/freshlien
cd freshlien && cp .env.example .env # fill in your keys

# 2. Backend
cd backend && pip install -r requirements.txt
uvicorn main:app --reload --port 8000

# 3. Frontend
cd frontend && npm install
npm run dev # localhost:5173

# 4. Scraper (one spider)
cd scraper && pip install -r requirements.txt
scrapy crawl nj_burlington -o output/nj_burlington.json
```

```
# 5. Deploy frontend
vercel --prod

# 6. Deploy backend (Render.com)
# Connect GitHub repo → set env vars → auto deploys on push
```

NEXT STEPS — HOW TO USE THIS DOCUMENT

You now have the complete master document for FreshLien.

Give this PDF to Cursor, Claude Code, or any AI IDE as context.

Start with Section 17 prompts — copy any prompt directly into your IDE.

Build in order: Schema → Scraper (NJ first) → FastAPI → React dashboard → Stripe → Launch.

freshlien.com

Same-day foreclosure intelligence — before the crowd

Prepared for Waqas Dost · Confidential · v1.0 · June 2026